

Logic Programming

Logic

- ▶ Logic is the study of the methods and principles used to distinguish correct from incorrect reasoning.
- ▶ It can be defined as study of principles of correct reasoning.
- ▶ The logic process takes in some information called premises and produces some outputs called conclusions.

Symbolic Logic

- ▶ Method of representing logical expressions through the use of symbols and variables, rather than in ordinary language.
- ▶ Its normally divided into two branches:
 - ▶ 1. Propositional Logic
 - ▶ 2. Predicate Logic

Propositional Logic

- ▶ Simplest form of formal logic.
- ▶ All statements made are called propositions.
- ▶ The FORMAL LOGIC is concerned with syntax of statements and not with their semantics.
- ▶ Propositional Logic takes only two values, either TRUE or FALSE.
- ▶ Example:
 - ▶ Sky is Blue
 - ▶ Pakistan is a city.

Syntax of Propositional Logic



1. Letters A, B, ..., Z and these letters with subscripted numerals are well-formed atomic propositions.
2. If A and B are well-formed propositions so are
 1. $\sim A$ (Negation of A)
 2. $(A \& B)$ (Conjunction of A with B)
 3. $(A \vee B)$ (Inclusive Disjunction of A with B)
 4. $(A \rightarrow B)$ (A implies B)
 5. $(A \leftrightarrow B)$ (Material biconditional of A with B)
 6. $(A \oplus B)$ (Exclusive Disjunction of A with B)
 7. $(A | B)$ (Joint Denial of A with B)
 8. $(A \downarrow B)$ (Disjoint Denial of A with B)

Semantics of Logical Proposition



Truth tables for Logical Connectives

A	B	$\sim A$	$\sim B$	$A \vee B$	$A \& B$	$A \rightarrow B$	$A \leftrightarrow B$	$A \oplus B$	$A B$	$A \downarrow B$
T	T	F	F	T	T	T	T	F	F	F
F	T	T	F	T	F	T	F	T	T	F
T	F	F	T	T	F	F	F	T	T	F
F	F	T	T	F	F	T	T	F	T	T

First-order Predicate Calculus

- ▶ First-order predicate calculus classifies the different parts of such statements as follows:
 1. Constants
 2. Predicates
 3. Functions
 4. Variables
 5. Connectives
 6. Quantifier
 7. Punctuation Symbols

Predicates

- ▶ These are names for functions that are true or false, like Boolean functions in a program.
- ▶ Defined as a relation that binds two atoms together.
- ▶ Example: Amit likes sweets. `likes(amit, areoplanes)` amit and sweets are atoms; likes is a predicate

Contants

- ▶ * These are usually numbers or names.* Sometimes they are called atoms, since they cannot be broken down into subparts.
- ▶ Example `natural(0)`.
- ▶ 0 is contant

Variable

- ▶ Variables Stands for Quantities that are yet Unspecified
- ▶ Example valuable(X)
- ▶ X is a variable.

Function

- ▶ Predicates-true or false and all other are functions which represent non-Boolean values

Unification

- ▶ Unification means making expression looks identical.
- ▶ It can be done by process of substitution.
- ▶ Unification Condition:
 - 1) Predicate symbol must be same.
 - 2) No. of argument in both expression must be identical
 - 3) If two similar variable present in same expression then unification fails.

Resolution

- ❑ Resolution means the fineness of detail that can be distinguish in an image or in a video display terminal.
- ❑ Example
 - a) Ravi likes all kind of food.
 - b) Apple and chicken are food.
 - c) Anything anyone eats and is not killed is food
 - d) Ajay eats peanuts and still alive
 - e) Rita eats that Ajay eats

Prove Ravi likes Peanut

$\neg$ likes(Ravi, Peanut)

Resolution

Steps:

- Negate the statement to be proved.
- Convert given facts into FOL.
- Convert FOL into CNF.
- Draw resolution Graph

$\forall x: \text{food}(x) \rightarrow \text{likes}(\text{Ravi}, x)$

1. $\text{Food}(\text{apple})$
2. $\text{Food}(\text{chicken})$
3. $\forall x \forall y: \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \neg \text{food}(y)$
4. $\text{Eats}(\text{Ajoy}, \text{peanuts}) \wedge \text{alive}(\text{Ajoy})$